Native Instance Ensembles with Expert-Aware Contour Refinement

Solar Filament Segmentation Challenge 2026

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ABSTRACT

We describe a trainable, image-only pipeline for segmenting solar filaments in native-resolution H-alpha observations. Seven YOLO segmenters and a Mask R-CNN contribute instance proposals; score-weighted mask voting is followed by three oriented-crop contour refiners. Alternative annotations are treated as expert-specific detection targets and conditional soft contour targets. This participant-trained pipeline achieved public panoptic quality (PQ) 0.39. A separate historical 0.57 entry depends on unavailable third-party detector weights and is not reproduced here. All eleven weights of the described pipeline are available locally, and a staged neural replay generated valid predictions for all 180 test observations. Controlled studies reject several plausible changes: larger classification loss, looser box suppression and a learned photometric selector. We distinguish measured validation, public score, replay fidelity and remaining release work.

Review candidate, November 6, 2026. This report describes the measured 0.39 pipeline. Final model selection, public artifact hosting and the organizer submission remain pending.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1]. Filaments are narrow, faint and frequently fragmented. A small contour change can move an instance across the IoU matching threshold. Experts also differ in whether a weak structure is present and whether nearby segments belong to one instance. These properties motivate preserving instance identities throughout the pipeline and separating detection from conditional boundary refinement.

Our deployment uses only the supplied H-alpha JPEG for each observation. It requires neither annotations nor additional observations, solar channels or adjacent frames at inference. The ensemble favors model and target diversity; the refiner improves boundaries without inventing new instance identities. The public result is modest relative to the current leader. Reproducibility and valid experimental selection are therefore necessary evidence, rather than evidence of a winning score.

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026 2026.

2 DATA, SUPERVISION AND EVALUATION

2.1 Official observations and expert targets

The official release contains 707 training photographs, 1,154 expert-image records and 8,199 annotated instances. There are 180 test photographs. Every image is 2048×2048. Chirality categories are mapped to one foreground class for segmentation. For the main YOLO members, each expert record supplies a separate training example. Instances from different experts are never simply concatenated, which would create overlapping duplicate labels. One diversity member uses geometrically merged expert instances instead. Mask R-CNN gives each physical photograph equal exposure and cycles its experts between epochs. These are distinct sampling objectives.

Two development cohorts must not be confused. The historical detector partition contains 565 training and 132 validation photographs, with ten additional photographs excluded because their capture dates overlap training. Its validation has 211 expert records. This partition inherited an upstream selection and is date-disjoint, but is not month-disjoint. Month-grouped confidence calibration cannot remove that training-level limitation. We therefore also ran a new whole-month split with 524 training photographs (861 records, 88 months) and 183 validation photographs (293 records, 30 months). The latter is used for the new 40-epoch classification-loss comparison. Neither cohort supplies an independent validation score for models refitted on all 707 photographs.

2.2 Permitted archive augmentation

The organizers permit additional unlabeled GONG frames and transfer of official training labels over short time offsets. All sites within an inclusive one-hour window around any test timestamp are excluded. We do not use the external MAGFiLO labels, SWEFil, HEK, new human annotations or test pseudo-labels. The one deployed neighbor-trained member used the recovered historical manifest: 2,683 transferred expert/frame rows corresponding to 1,649 distinct same-site frames, at offsets of 60 or 120 seconds. Its annotation source is official training data, and every timestamp passes the test exclusion.

Separately, an audit covers all 2,696 cached external frames, including 874 registered cross-site frames. None violates the test exclusion. Registration and availability do not establish a model gain; the cross-site cache is not used by the eleven deployed weights. A later correlation audit rejects 76 poor-quality same-site frames. Of these, 68 occurred in the historical neighbor fit. This is a documented quality limitation, not an exclusion-window violation. New experiments apply the reviewed quality filter and the development partition guard.

Table 1: The eleven weights of the measured public 0.39 pipeline. Counts denote physical training photographs; NB denotes the historical short-offset neighbor manifest. All-data fits have no independent held-out score.

Model	Training targets	Input
YOLOv8l, long	565, expert records	2048
YOLOv8l, NB	707 + transferred records	2048
YOLOv8l, deployment	707, expert records	1536
YOLOv8l, two seeds	565, expert records	1536
YOLOv8l, union	565, merged experts	1536
YOLOv8x	565, expert records	1536
Mask R-CNN R50-FPN V2	707, cycling experts	2048
U-Net/B0, two seeds	565 then 707, soft contours	512×128
U-Net/ResNet34	565, soft contours	512×128

2.3 Metric and selection

We use the host's pooled all-pair metric over expert-image records:

$$\mathrm{PQ} = \frac{\sum_{(g,p):\mathrm{IoU}(g,p)>0.5} \mathrm{IoU}(g,p)}{\mathrm{TP} + \frac{1}{2}\mathrm{FP} + \frac{1}{2}\mathrm{FN}}.$$

All qualifying pairs contribute; a Hungarian assignment is not substituted. Predictions are made pixel-disjoint before scoring. Confidence thresholds are selected outside each reported fold, grouping photographs by calendar month. Paired confidence intervals resample whole months. These intervals describe the particular validation cohort; they do not estimate the hidden leaderboard score.

3 INSTANCE PROPOSAL ENSEMBLE

3.1 Members and training

Table 1 lists all deployed weights. YOLO uses generic COCO initialization and Ultralytics 8.4.183 [6]. The principal recipe uses AdamW, learning rate 10^{-3}, cosine decay to one tenth, nominal batch 16 with physical batch one, weight decay 5×10^{-4}, non-overlapping mask targets and mask ratio four. Rotation spans 180°; each flip has probability 0.5. Translation and scale ranges are 0.05 and 0.15, and value jitter is 0.15. Mosaic, mixup and copy-paste are disabled. Full protocols record seeds, actual checkpoints and the original time-limited schedules. For example, the long 2048 member completed 76 epochs under a nominal 70-epoch, 9.5-hour schedule. A time cap is not an exact epoch guarantee.

The native Mask R-CNN [3, 5] retains its generic COCO backbone, FPN, proposal network and intermediate heads. Only the final task predictors are replaced. Mask ROI pooling is 56×56 and deconvolution gives 112×112 mask logits. Training uses binary cross-entropy plus soft Dice for positive ROIs, alongside the standard proposal and box losses. The deployment fit contains 60 epochs and 10,605 successful optimizer updates, physical batch one with accumulation four, SGD learning rate 0.005, momentum 0.9, gradient clipping five, 500 warmup updates and cosine decay. Geometry augmentation uses exact quarter turns and flips; intensity augmentation varies gamma, contrast and gain. Checkpoint provenance binds the recipe, source and official annotation hashes.

3.2 Image normalization and proposal fusion

Each YOLO member uses per-image BatchNorm statistics. BatchNorm modules run in training mode with zero running-statistic momentum; convolution and BatchNorm fusion is disabled. Thus a prediction depends on its own image, while stored running statistics do not change. Candidate export uses confidence 0.05, box-NMS IoU zero and at most 100 instances. The 1536 members restore masks at native resolution. Mask R-CNN uses four flipped views whose predictions are transformed back before fusion.

Proposals are greedily clustered at mask IoU at least 0.5, allowing at most one contribution from each detector. Within a cluster, foreground pixels are retained when their score-weighted vote reaches half of the total weight. Confidence is the summed member score divided by eight, so a missing member contributes zero. At confidence 0.2, score-ordered ownership resolves overlap and masks smaller than five pixels are discarded. This is instance fusion, rather than averaging a semantic foreground image and recovering instances afterward.

The implementation votes only on the nonzero native-pixel supports. Its float32 reduction order matches the original dense computation. All 3,457 archived test clusters and the resulting 1,270 seeds reproduce the dense masks exactly. The measured fusion/ownership stage takes 149 seconds for 180 photographs; this is not the full neural runtime.

4 CONDITIONAL CONTOUR REFINEMENT

For each seed, its principal orientation defines a 512×128 crop. Inputs are normalized grayscale intensity and the seed mask. Training seeds are selected official expert instances, not predictions from the unavailable third-party detector. For each seed, the most overlapping instance from each expert contributes only if IoU exceeds 0.15. Their mean warped masks form a soft target conditional on the structure being annotated. This avoids treating every omitted weak filament as an incorrect contour. The initial crop cache has 4,147 instances from the 565 training photographs.

U-Nets use EfficientNet-B0 and ResNet34 encoders from Segmentation Models PyTorch [4]. The original 565-image B0 checkpoint is continued for 3,000 updates on all 707 photographs, independently with seeds 20261022 and 20261023, at maximum learning rate 5×10^{-5}. ResNet34 retains its 565-image fit. The protocol labels of the B0 continuation refer to an older detector used for export; the training crop audit confirms that no private detector is needed to construct its supervision.

At inference, four crop flips are averaged for each refiner. Quantized probabilities from the B0 pair are averaged in the original order; the final mixture assigns weight 0.75 to this pair and 0.25 to ResNet34. Contours are thresholded at 0.5 with a 24-pixel growth cap. If refined/seed IoU is below 0.2, the seed is retained. A half-weight signed-distance blend with the original contour limits boundary drift; detached components below 5% of the largest component are pruned. Original instance identities and score-ordered pixel ownership persist throughout. These deployment settings are historical selections; the full-data refiners are not independently validated by their own training photographs.

Table 2: Recent measured comparisons. A: 132 photographs/211 expert records; B: 183 photographs/293 records. Threshold calibration is outside reported folds. Intervals resample months.

Comparison	Reference PQ	Change
A, five-bank suppression retuning	.41998	−.00165
A, box NMS 0.7 with mask deduplication	.42057	−.00036
B, YOLO11m classification 1.5 vs. 0.5	.38272	−.00914
B, nested photometric selector	.38272	−.00178

5 MEASURED STUDIES AND FAILURE ANALYSIS

The recent studies in Table 2 use frozen candidate banks or controlled neural fits. Their absolute scores should not be ranked across different cohorts or model families. None of the rejected changes was promoted to a new competition submission.

On the five-bank development ensemble, a fixed duplicate-removal setting increases PQ from 0.419981 to 0.420757, removing five false positives without losing a true positive. Its paired interval runs from 0.000142 to 0.001711. However, selecting suppression settings outside each fold yields 0.418333, a change of −0.001648 with interval -0.007937 to 0.004652. The favorable fixed setting was therefore not accepted as a robust deployment gain. An error audit attributes 594 misses to confidence (221), inaccurate extent (187), no nearby raw proposal (120), fusion (64), and final ownership (2). This diagnosis motivates better proposal recall and mask extent, rather than relying on cosmetic postprocessing alone.

Relaxing box NMS produces 2,367 rather than 1,209 raw proposals, but the calibrated score changes by only −0.000357, with interval -0.006511 to 0.005212. Without mask deduplication, its fixed-threshold PQ falls to about 0.393. More proposals do not by themselves establish better filament recall at the chosen false-positive rate.

For the whole-month experiment, two YOLO11m/COCO arms share initial parameters, data, augmentation traces, seed, 1280 resolution and 40 epochs. Both consume 8,640 batches and complete 622 updates with no scaler skips. Classification weight 0.5 scores 0.382718; weight 1.5 scores 0.373576. The paired difference is −0.009142, with interval from -0.017412 to -0.000885. Increasing this loss is rejected. This is a controlled smaller-model experiment, not a reconstruction of the private native-resolution detector.

A separate 14-feature photometric/geometry head uses image contrast, seed shape, position and detector confidence. An inner three-month-fold loop selects its threshold; an outer five-month-grouped loop reports performance. It scores 0.380943 against 0.382718, difference −0.001775 with interval from -0.006497 to 0.002650. Removing more false positives also removes true positives, so this selector is rejected. No test images were used to fit or choose the head.

6 REPRODUCTION, LIMITATIONS AND RELEASE

The accepted participant-trained entry scored public PQ 0.39 (submission 57732339). Its CSV contains 1,270 disjoint native instances and explicit inference records for all 180 observations. The separate historical public 0.57 entry uses an unavailable private detector. Public-source research recovered its inference settings, but neither its weights nor its training recipe. We do not label an independently trained YOLO model as those original weights.

The local release bundle contains all eleven described checkpoints, 1,013,552,336 bytes in total, plus a model/source SHA-256 manifest. A fresh staged replay runs the eight detectors, then the three refiners, from the official test JPEGs. It produces 1,270 valid masks over all 180 observations. Agreement PQ with the accepted 0.39 predictions is 0.997954; 83 observations have changed mask pixels. This is replay fidelity, not segmentation accuracy. The newly computed CSV has not been submitted or scored. A two-image probe of the faster integrated runner matches that staged replay exactly; a complete notebook execution remains a separate release gate.

The verified inference environment uses Python 3.12, PyTorch 2.11.0+cu128, Torchvision 0.26.0+cu128, Ultralytics 8.4.183, Segmentation Models PyTorch 0.5.0 and timm 1.0.31. GPU refinement uses float32 with autocast and TF32 disabled. Source and package pins are recorded; hardware-independent byte identity is not asserted. Model validation, output geometry, temporal validity and license obligations are distinct checks. Public hosting must allow the demo to fetch and verify weights automatically. The complete training walkthrough, public release, final model choice and organizer form are still being prepared. No prize-eligibility certification is claimed.

The principal scientific limitations are historical partition bias, annotation disagreement, public-board selection exposure and the lack of independent validation for all-data refits. Grouped holdouts and rejected negative studies make these limits visible. Future changes must improve held-out proposal recall or contour extent under the same evaluation protocol before replacing the deployment model. AI assistance was used for research and implementation; the participant remains responsible for understanding and verifying the work.

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